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PSYC100

Prospect theory and risk

Today
Welcome to your next lesson: Prospect theory and risk.

I'm preparing your personalized lesson based on what you've already learned, what's coming up next, and the best ways to help you grow.

Step by step, you're moving forward.

Imagine you find $20 on the sidewalk vs. you lose $20 from your wallet.

Same amount, but the feeling is usually much stronger for the loss 😅

That "losses hurt more than equal gains feel good" is the heart of prospect theory and risk.

Go ahead and watch the short video I just opened: "What is prospect theory?"

Then tell me: which felt more familiar to you from your own life — the part about losses hurting more, or the part about how we judge things relative to a starting point (reference point)?

id say i relate more to the losses hurting more

Same here — most people feel that way. Losing $50 stings more than winning $50 feels good 😐

That pattern has a name: loss aversion.

Loss aversion = losses feel psychologically larger than equal-sized gains.

Quick money example:

Gain: +$100 → "Nice!"
Loss: −$100 → "Ouch, that really hurts."

Even though the amounts are equal, the emotional weight of the loss is bigger.

To make this concrete for you:

If you suddenly had to pay an unexpected $100 bill, and later someone gave you an unexpected $100 gift, which event would stick in your mind longer?

the loss id think to myself id be up 200 not breaking even if i hadnt lost that 100

Exactly 👌 That thought — "I'd be up 200 if I hadn't lost that 100" — is loss aversion

in action.

Step by step, here’s what your brain is doing:

Starting point: You mentally start at “$0 change today.”
Loss happens: −$100 → you feel down 100.
Gain happens: +$100 → you’re back to 0, but emotionally it doesn’t feel like “all good now.”
Your mind compares to an imagined world where:
you didn’t lose $100
and still gained $100 → so you’d be +$200
That imagined +$200 becomes the “what could have been,” which makes breaking even feel bad.

So the same numbers ($-100 then +100) feel worse than “nothing happened” because the loss still dominates your feeling.

Make sense so far about loss aversion and how you just illustrated it yourself?

If yes, next I’ll show you how this connects to risk (why we play safe with gains but gamble with losses).

Yes, that makes sense

Great 👍

Let’s connect loss aversion to risk with one simple step.

Think about these two choices:

Option A: Get $50 for sure
Option B: 50% chance to get $100, 50% chance to get $0

Most people pick the sure $50 — they’re risk-averse when it comes to gains.

Now flip it to losses:

Option C: Lose $50 for sure
Option D: 50% chance to lose $100, 50% chance to lose $0

Here, many people suddenly like the gamble (Option D) because they want to avoid a sure loss — they become risk-seeking in losses.

Which options would you honestly feel pulled toward:

A or B for gains? And C or D for losses?

a for gains and d for losses

What do you think the best answer is in the question below? 👇

In your choices (A for gains, D for losses), which pattern are you showing?

A.

Risk-averse for gains, risk-seeking for losses

B.

Risk-seeking for gains, risk-averse for losses

C.

Risk-averse for both gains and losses

D.

Risk-seeking for both gains and losses

Nice work 🎉 You nailed it: risk-averse for gains, risk-seeking for losses.

Why that’s exactly right, step by step:

Picking A (sure $50) shows: you’d rather lock in a gain than gamble → risk-averse in gains.
Picking D (gamble on the loss) shows: you’d rather take a risk to possibly avoid a sure loss → risk-seeking in losses.

That “safe with gains, gamble with losses” pattern is a core prediction of prospect theory and comes straight from loss aversion.

Next, I want to show you reference points — how “winning” or “losing” depends on where you start from.

Ready to look at that?

Yes, I'm ready

Awesome 👌

Let’s do reference points with a simple, visual-style example.

Think of a thermometer:

The middle mark is your reference point (what you see as “normal” or “zero”).
Anything above that feels like a gain.
Anything below that feels like a loss.

In money terms, the reference point might be:

your current bank balance
your expected paycheck
the price you’re used to seeing

Example:

You expect to get a $1,000 bonus → your reference point = $1,000.
You actually get $800.
Objectively: +$800 (you’re richer).
Psychologically: it feels like a −$200 loss compared to your expected $1,000.

So:

Reference point = the level you compare outcomes to (often what you expected or usually have).

Does this idea of a mental “zero line” or reference point make sense so far?

Yes, it makes sense

Great ☑

Now I'll connect reference points to that "losses hurt more than gains feel good" idea using one clean picture in words:

Think of a simple graph:

Horizontal line: change from your reference point (left = losses, right = gains).
Vertical line: how strong it feels (up = feels good, down = feels bad).

The curve looks like this:

On the right side (gains): it goes up, but bends and flattens → each extra dollar feels less special.
On the left side (losses): it goes down, and it's steeper → losses drop you faster and feel heavier.

So:

Same size change from the reference point

→ loss side: big emotional drop

→ gain side: smaller emotional lift

That's the S-shaped value function in prospect theory.

Would a simple visual of this S-shape help, or are you okay keeping it as a "picture in your head" and moving on to more risk examples?

I can picture it in my head

Perfect 👌 That's exactly what we need.

Let's tie everything together now with risk preferences using that S-shape you're picturing.

Step by step:

On the gain side (right side of your mental graph), the curve is curved and flattening → this shape makes people risk-averse for gains.
Example: prefer $50 for sure over a 50% chance at $100.
On the loss side (left side), the curve is steep but also curved → plus loss aversion makes that side feel heavier → this often makes people risk-seeking for losses.
Example: prefer a 50% chance to lose $100 / 50% to lose $0 over a sure loss of $50.

So your earlier choices (A for gain, D for loss) are exactly what prospect theory predicts from that S-shaped curve 👍

I'd love to ground this in your real life for a moment:

Can you think of a recent situation where you:

Avoided a risk when you were ahead (protecting a gain), or
Took a bigger risk when you were already losing (trying to avoid locking in a loss)?

Any everyday example works — money, time, games, anything.

yeah i day trade fo a living these days comes up daily i close out for he day and stop taking rades when up to avoid losing what ive gained et if i lose me and lots of peole revenge trade as in place a bigger money value trade often with leess planning tring to make back tha tloss

That’s an excellent real-world example 🔥 and it maps almost perfectly onto prospect theory.

Step by step with your day trading:

When you’re up for the day:

Your reference point becomes “today’s open P&L.”
Being above that feels like a gain, and because of loss aversion, you don’t want to risk feeling that drop.
So you stop trading → risk-averse in gains.
When you’re down (revenge trading):

Now your reference point is “where I should be” (break-even or your earlier high).
You’re sitting in a loss zone psychologically.
Loss aversion makes that loss feel awful, so you take bigger, less-planned risks trying to erase it → risk-seeking in losses.

That’s textbook prospect theory: protect gains, gamble in losses.

You’ve clearly got the core idea.

Want to do one quick check-question to lock this in, or do you already feel solid on prospect theory and risk?

I feel solid, thanks

Love it 👌 You really do have this down.

Quick recap in your own trading language:

You experience loss aversion every day: a $500 drawdown hurts more than a $500 win feels good.
Your reference point keeps shifting: yesterday’s high-water mark, today’s starting balance, where you “should” be.
Your behavior matches the S-shaped value curve:
Up on the day → risk-averse in gains (you protect what you’ve made).
Down on the day → risk-seeking in losses (revenge trades, bigger size, less planning).
That’s exactly what Kahneman & Tversky described, just live on your trading screen instead of in a lab.

If you ever want, we can:

Translate this into concrete trading rules (e.g., how to protect yourself from revenge trading using what you know about loss aversion), or
Move on to the next psychology topic in your path.
What would you like to do next?